



IIIT-Pune

RiskSignal Engine

Module-B

Explainable AI/NLP Risk Signals from news and social text, with Event-driven
portfolio stress testing

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Problem and Approach

Problems

Unstructured Global Noise

Market-moving events are buried in large volumes of news and social media, making relevant signals difficult to identify.

Monitoring Gap

Manual monitoring is slow and fragmented, causing delayed detection of emerging risks across companies and sectors.

No Actionable Risk Context

Raw headlines lack structured impact scores, portfolio-level exposure analysis, and explainable stress scenarios

Solution

Collect

Ingest news and social signals from GDELT, Bluesky and replay data

Interpret

Use NLP to resolve entities, analyze sentiment and classify risk events.

Quantify

Generate explainable impact scores and run portfolio stress scenarios.

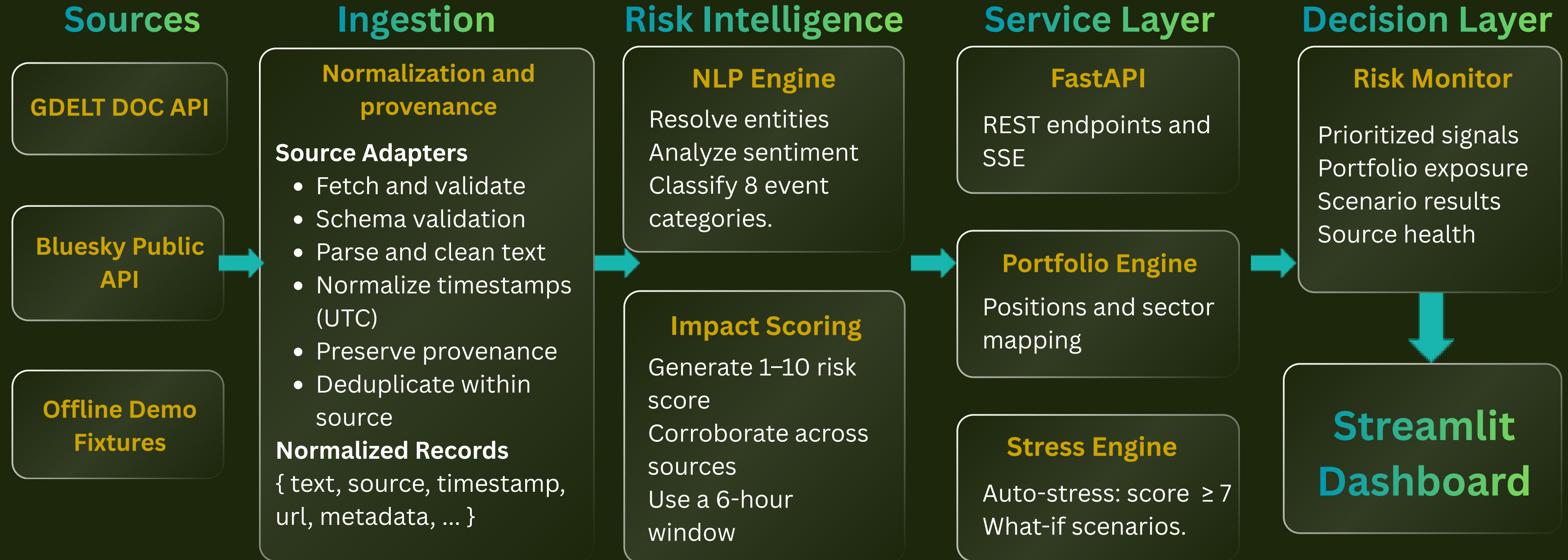
Act

Present risk signals, scenario results and source health through a dashboard.



System Design

Architecture Diagrams & Data Flow



SQLite persists ingestion runs, risk signals and stress decisions



Technology Stack



Python
(Core pipeline)



Hugging Face
(NLP models)



SQLite
(Data storage)



FastAPI
(API & services)



Streamlit
(Dashboard)



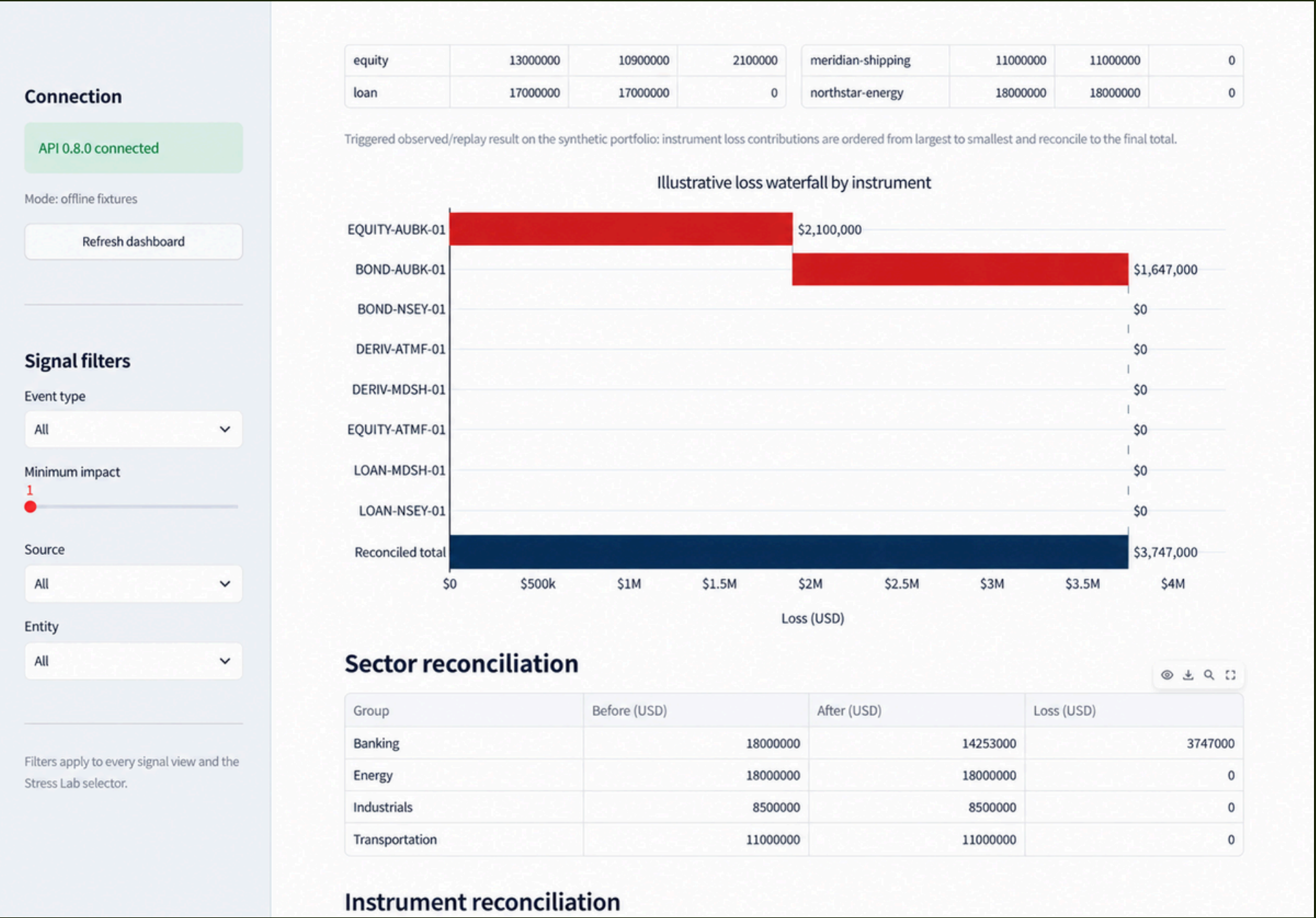
Implementation Highlights

MODULE	CHOICE	WHY?
INGESTION	GDELT and Bluesky adapters, checksummed fixtures and replay	Two source types; isolated failures; works offline
NLP ENGINE	FinBERT and MiniLM, plus transparent rule baseline	Pinned models; explicit mode, no silent fallback
IMPACT SCORE	Weighted formula over 6 stored factors	Auditable; direction-agnostic and disclosed
STRESS ENGINE	Decimal valuation for bonds, loans, equities, derivatives	8 event scenarios; reconciles to the cent
API AND UI	FastAPI (10 endpoints), SQLite, Streamlit	Typed contracts; simple local run



Key Results

MEASURE	NAIVE BASELINE	RESULT
Sentiment accuracy, Financial PhraseBank (2,264 rows)	Rule lexicon: 0.677	FinBERT: 0.972*
Sentiment macro-F1	0.430	0.963*
Organic stress trigger	Fixtures: none above 7	Replay: 4 of 4 score 9
Replay scenario loss	n/a	USD 3.747M of 55.50M; reconciliation 0.00
Quality gates	n/a	133 tests; 9/9 checksums; validator 0.07 s



*Four fictional impact-9 signals produced an illustrative \$3.747M Aurora Bank portfolio loss, reconciled to \$0.00.



Domain Impact

Why the Solution Matters

- Transforms large volumes of unstructured financial news into actionable risk intelligence.
- Enables early identification of emerging market risks and potential portfolio exposure.
- Reduces manual effort in monitoring news and assessing financial impact.

Use-Case Impact

- Automatically detects, classifies, and prioritizes market-moving events.
- Links relevant events to affected sectors and portfolio positions.
- Provides scenario-based stress estimates and visual dashboards for faster analysis.

Business Impact

- **Faster Decisions:** Quickly identifies critical market events.
- **Risk Visibility:** Highlights potential portfolio risks.
- **Efficiency:** Automates news monitoring and risk analysis.
- **Scalability:** Supports future expansion of risk monitoring.
- **Auditable by design:** Scores and skipped decisions track sources, factors, and model versions.



Limitations and Future Scope

Limitations:

- **Synthetic Data Dependency:** Portfolio, scenarios, and replay datasets are synthetic; live sources depend on external APIs and availability.
- **NLP & Scoring Accuracy:** Rule-based classification and heuristic impact scores lack comprehensive real-world validation.
- **Simplified Stress Modelling:** Valuation excludes advanced factors such as liquidity, convexity, optionality, and basis risk.
- **Restricted Scope & Testing:** Live integrations are mocked; real trading, investment recommendations, and production deployment are excluded.

Future Scope:

- **Real-World Data Integration:** Incorporate licensed datasets and human-labeled events for reliable evaluation.
- **Advanced AI & NLP:** Benchmark and calibrate FinBERT, event classification, severity estimation, and impact scoring.
- **Enhanced Stress Testing:** Introduce advanced valuation models, liquidity risk, options, and dynamic market scenarios.
- **Production-Ready Architecture:** Implement scalable databases, secure authentication, audit trails, cloud deployment, and durable event streaming.